Calc Proofs

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Abstract

Yeah, this is where I'll put my proofs in the future for safekeeping so I can reference them later or smth.

1 Limits

Limit proofs hinge on the following definition of a limit:

$$\lim_{x \to a} f(a) = L \iff \forall \varepsilon > 0,$$
$$\exists\, \delta \; s.t. \; 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon$$

1.1 Addition Rule

Statement:
given that

$$\lim_{x \to a} f(x) = L, \lim_{x \to a} f(x) = M$$

, show that

$$\lim_{x \to a} f(x) + g(x) = L + M$$

Proof. Consider some $\varepsilon > 0$, then by the definition of the limit there exists δ_1 and δ_2 such that:

$$0 < |x-a| < \delta_1 \implies |f(x) - L| < \frac{\varepsilon}{2}$$

$$0 < |x-a| < \delta_2 \implies |g(x) - M| < \frac{\varepsilon}{2}$$

Then let $\delta = \min(\delta_1, \delta_2)$, and we need

$$|f(x) + g(x) - (L + M)| < \varepsilon$$

We have:

$$0 < |x-a| < \delta$$

which implies

$$|f(x) - L| < \frac{\varepsilon}{2}$$

and

$$|g(x) - M| < \frac{\varepsilon}{2}$$

By the triangle inequality,

$$|a + b| \leq |a| + |b|$$

so

$$|f(x) + g(x) - (L + M)| < |f(x) - L| + |g(x) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

so

$$|f(x) + g(x) - (L + M)| < \varepsilon$$

and so limits follow the addition rule. ■ ■

Do note that I *did* need to consult the textbook for the triangle inequality stuff, but the rest was fine.

1.2 Multiplication Rule

Given that

$$\lim_{x \to a} f(x) = L$$

and

$$\lim_{x \to a} g(x) = M$$

prove that

$$\lim_{x \to a} f(x)g(x) = LM$$

Proof. ■

1.3 Polynomials are continuous

A function, $f(x)$, is continuous if

$$\lim_{x \to a} f(x) = f(a)$$

From this, let us prove that all polynomials of the form x^n for $n \in \mathbb{N}$ are continuous.